Assignment 2: Systemic Games with Emergent Mechanics

By Jake Kielty

Concept: Bombs and Fuses

The basic idea here is that the player can place a bomb that will detonate when the “signal” from a burning fuse reaches it.

1. The player first hits a button to place a bomb.
2. Now they will have to place the fuse (basically walking the path that they want it to take), and will press the same button they initially pressed to place the bomb to stop placing the fuse.
3. From there, the bomb is placed but inert and has to be lit. When a source of fire, either an explosion from another bomb or a fire arrow, hits the un-lit fuse, it is ignited.
4. While the fuse is lit, the fire will slowly travel down the length of the fuse towards the bomb, removing sections of fuse as they are burned up.
5. When it reaches the bomb, KABOOM!
6. The explosion can light fuses, destroy objects/terrain, and/or damage enemies and the player.

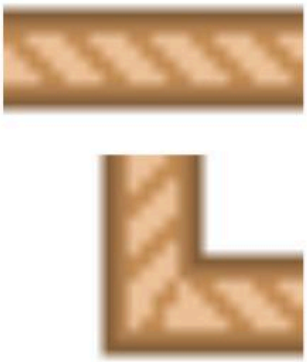
The bomb could be used for combat, for removing obstacles, or for opening secrets/passages (Similar to existing Zelda bombs, these are basically just time-delayed/remotely triggered). The bomb could even be set off by non-player sources, or could be used by other things besides just the player. These could even be pre-placed in the level and can be used in creative ways to create challenges (ex: the player has to race the fuse to the end of a maze-like room before it reaches the bomb, causes a cave in, and traps the player there forever).

Object: Bomb

Unlit	Lit

Properties	• Toggle State: Unlit or Lit • Color: Grey or Red
Behaviors	• When a **lit** fuse reaches the bomb, it will turn red for five seconds before exploding, dealing damage in a radius around it, and destroying any destructible objects/terrain. • Explosive (Can destroy things)
Game State Feedback	Visual: • Changes color to appropriate color. • Smoke particles appear over the bomb when it is lit. Audio: • Plays sound when lit to indicate it's 5 seconds from detonation.
Systemic Feedback	• Explosion can affect other objects. (ex. The explosion can light other fuses, can open new pathways, etc.)

Object: Fuse

Unlit	Lit
	

Properties	• Toggle State: Unlit or Lit • Color: Beige/Brown or Red
Behaviors	When lit by a source of fire (explosion, fire arrow, etc.), the currently burning section turns red. After 1-2 seconds (this would be fine-tuned), the currently burning section disappears and the next connected section of fuse changes color and begins burning like the last. This is how the signal "travels" towards the bomb.
Game State Feedback	Visual: • Changes color to appropriate color. • Smoke particles appear over the lit section of fuse. Audio: • A constant "hissing" sound is played when a fuse is burning to indicate that something is going on.
Systemic Feedback	Passes "signal" to next fuse. Can be ignited by two things: A bomb's explosion, and a Fire Arrow.

Object: Fire Arrow

Properties	• Simple projectile. • "isFire"?? (If it has any variables, it would be one that distinguishes it from the regular projectiles/arrows)
Behaviors	When a fuse is hit by a source of fire, either an explosion or by one of these arrows, it will ignite and begin burning down. (Presumably, these could also serve as a tool, and could be used to fight enemies, burn away debris obstructing the player, or ignite other objects to solve puzzles or navigate)
Game State Feedback	Audio: • Fire Arrows have a different sound than regular arrows, with a crackling burning sound that plays alongside the regular arrow firing sound.
Systemic Feedback	Ignites flammable objects when it hits them. In this case, it ignites fuses when it hits them.

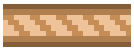
Components

The idea with the bomb here is to provide the player with a mechanic that can serve as a combat tool, a puzzle-solving tool, and a possible level challenge. There are three main components in this design that make it work: The bomb, the fuse, and the fire arrow. Here are each of them in more detail:

The Bomb

The Bomb is the “business” part of this system, and is the part that affects the game the most directly. When the fuse reaches the bomb, after five seconds, it explodes, and deals damage to anything around it, including the player, enemies, and objects/terrain, meaning it can be used in combat with careful preparation, and in solving puzzles or navigating with how it could be used to open secret passageways or ignite *other* bombs. The player does not interact with it directly once placed, but interacts with it through the fuse.

The Fuse



The Fuse is the part that makes the bomb actually explode, and it is the part that makes this more than the basic Zelda bomb item. The idea here is that players must light the fuse somehow if they want to set off their bombs, and so they must consider where they route this to, and if they can hit it with something that will ignite it. It can also have a maximum range if length ends up being an issue.

The Fire Arrow

The Arrow is the part that allows the player to directly interact with the fuses, as otherwise, the only way to light them would be to have another bomb light them. With this, the player can have the ability to light their fuses at a distance, which can be used to make puzzles and combat challenges.

Justification

This is, overall, a fairly straightforward and simple game mechanic, but one that should allow for some interesting possibilities, both for the designer and for the player. The basic idea here is that the player will be able to use this to solve problems, find secrets and/or shortcuts, and fight enemies.

With regards to the bomb in particular, the idea is that you've got the classic Zelda bomb, where the player can place it and detonate it, but with the added condition that it must be lit first rather than it just being automatically lit when you deploy it. This forces the player to more strongly consider where they place the bomb (considering now that they have to have a way to light it), and *why* they are placing it, as it's more of an effort to detonate it now that you've got to run a fuse and everything.

This will also allow for some more interesting strategic opportunities in combat for more patient players, as this bomb can also be used to set up a trap for enemies. Depending on how exactly the fuse moves, and how fast it is, one could place it in an area, run a fuse, and then wait for the right timing for when enemies move past it to set it off (this would be a more valuable strategy if the bombs were highly potent against enemies, as it may not be worth all of the setup if it's not doing substantial damage).

As for the fuse, this is basically just a means to an end as far as objects go, and while it's the critical part that makes the bombs blow up, it's not particularly interesting by itself. The general idea here is that, like a real fuse, once lit, it sends a signal down the line to a bomb of some variety. Depending on how the game would actually turn out, putting the specifics of how exactly it would burn out aside, parts of the fuse would get burned out as it goes, and would disappear. This serves as a visual timer and indicator that a) the clock is ticking, and b) something is going to happen when that fuse runs out.

Finally, for the Fire Arrow, this one has much potential beyond this idea with the bombs, and could be tied into a broader combat system with fire, the ability to ignite objects/enemies/terrain or other things. This one is simple, but serves as a means by which the player can *directly* light the fuses themselves. Without this, players are limited to other bombs as a means of igniting their own, which would make the bombs far less useful. They could be unlimited, or could be a limited resource, depending on how the bombs fit in with the broader combat sandbox.

While regular bombs work great, the ability to have a fuse allows for remote detonation, which enables a range of other actions for the player, both in and out of combat. Overall, these mechanics are simple, but could easily allow for some interesting puzzle-solving and combat possibilities.